

Chinmay Bhansali

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Research Interests

Core Areas: Human-Computer Interaction, Human-AI Interaction, Computational Design Evaluation

Focus: Evaluation methods for AI-mediated design workflows, cognitive misalignment in AI-generated interfaces, how non-UX professionals make interface decisions using generative tools

Education

University of British Columbia

Vancouver, BC

Bachelor of Science in Integrated Sciences

September 2022 – May 2026

Concentrations: Computer Science, Psychology, Mathematics (HCI)

Awards: Swami Vivekanand Scholarship for Academic Excellence (full tuition coverage for 4 years)

Research Experience

UBC Sensory Perception & Interaction (SPIN) Lab – Research Assistant

October 2025 – Present

Working with Rocco Ruan (PhD student) on “Play Together, Feel Together,” investigating technology-mediated social touch in multiplayer games using Sony TOIO robots.

- **Prototyping:** Developing Unity-based multiplayer game prototypes incorporating Sony TOIO robots to facilitate remote social touch interactions, implementing new game features and mechanics to explore design variations for user studies.
- **User Study Support:** Co-designing and conducting user studies to evaluate how people respond to different technology-mediated touch designs, including developing study protocols, creating psychometric questionnaires, and performing thematic analysis on qualitative participant feedback.
- **Data Analysis:** Processing and analyzing movement trajectory and physiological data from experiments using Python, applying signal processing techniques including filtering, dynamic time warping, and cross-correlation analysis to identify behavioral patterns and meaningful interaction signatures.
- Contributing to research addressing loneliness as a public health issue by exploring how Sony TOIO robots can provide embodied representations of remote social touch in naturalistic gaming contexts.

Selected Writing

“The Trail Nobody Asked For” – LinkedIn Article

April 2026

Published article examining how AI-generated interfaces encode design choices that users never requested, introducing the friction tax: the accumulated usability cost users pay when AI-generated interface decisions diverge from their mental models. Analyzed a usability incident during a course-based HCI project in which an AI-built prototype added an unrequested visual drag trail that a trained UX team failed to catch before reaching users.

Industry Experience

Datamatics – Pulse J2EE Application Management

June 2023 – August 2023

Developed an enterprise-level web application for managing bulk employee data

- Modeled dynamic web pages using Struts and JavaScript, enabling efficient bulk data management
- Designed and optimized MySQL queries and stored procedures, ensuring robust data operations
- Deployed and configured application modules on Tomcat server, ensuring reliable web service delivery
- Implemented responsive user interfaces with HTML, enhancing user experience across devices

Selected Projects

GoodTimers – Advanced HCI Research Project (CS444)

January 2026 – April 2026

Researcher and Designer. Full HCI research pipeline from need finding through controlled experiment, designing and evaluating a mobile application for planning between-lecture walks at UBC. Received two CPSC 444 showcase awards: Best at Catching “Sneaky Factors” and Most Comprehensive Construction (the only team to receive two awards).

- **Need Finding & Design:** Conducted field study identifying that student break dissatisfaction stemmed from planning overhead rather than lack of activities; contributed to design requirements and evaluation of four design alternatives before converging on a walk-route generation approach.
- **Cognitive Walkthrough:** Participated in collaborative cognitive walkthrough of low-fidelity prototype, identifying confirmation and proceeding actions as the primary weak point in the design’s feedback structure; findings directly shaped the medium-fidelity prototype scope.
- **Prototyping:** Designed Figma mockups and built the medium-fidelity web prototypes (HTML, CSS, JavaScript; Webstorm with Gemini) used in the actual experiment after Figma proved insufficient for the drag interaction; during implementation, the AI-assisted coding produced a visual drag trail not present in the original Figma design, which generated user confusion in the pilot study and required redesign.
- **Experiment Design & Data Collection:** Co-designed a 3×2 split-plot controlled study (N = 12) comparing three decision-factor collection modalities against a Google Maps baseline; authored apparatus specification, questionnaire design, and study protocol; conducted study sessions with participants including post-study semi-structured interviews.
- **Analysis & Findings:** Team applied ART ANOVA for non-parametric factorial analysis, Wilcoxon signed-rank tests, paired t-tests, and Hedge’s g for effect size estimation; PDS significantly reduced cognitive load relative to the baseline (partial $\eta^2 = 0.69$, large effect); mixed-methods analysis combining quantitative measures with qualitative interview themes proved essential given N = 12 statistical power limitations.

Food Manager – HCI Research Project (CS344)

September 2025 – December 2025

Lead Researcher and Qualitative Data Analyst. Designed and evaluated a mobile application for reducing household food waste through a full user-centered design process. Received the People’s Choice Award.

- **User Research:** Conducted formative research (N = 11) using semi-structured interviews and affinity diagramming to identify user needs regarding household food waste and inventory tracking.
- **Iterative Design:** Piloted low-fidelity prototypes to validate interaction flows, utilizing findings to evolve the conceptual model from a Conveyor to a Factory metaphor before refining the final design into a medium-fidelity Figma prototype.
- **Usability Testing:** Designed and executed a mixed-methods summative evaluation (N = 11) utilizing think-aloud protocols and post-study questionnaires to assess interface efficiency, visibility, and integration.
- **Data Analysis:** Performed mixed-methods analysis by triangulating quantitative metrics including click-counts and Likert scales with qualitative insights from thematic analysis and observation notes to diagnose interaction bottlenecks.

ASLingo – ASL Learning Platform (Hackathon Team Project)

November 2023

Built a real-time American Sign Language learning platform in 36 hours using TensorFlow, MediaPipe, Flask, and Bootstrap.

- Integrating the gesture recognition pipeline made concrete a recurring design principle: classification latency of a few hundred milliseconds shifted user perception of feedback from response to failure, making an engineering constraint into an interaction design variable.

Technical Skills

Languages: C/C++, Java, C#, Python, R, SQL, HTML, CSS, JavaScript

Tools & Environments: Figma, Git, GitHub, Unity, Google Colab, Jupyter Notebook

Frameworks: TensorFlow, Flask, Struts, Tomcat, Bootstrap, MediaPipe, Entities-Components-Systems (ECS)